State of the Art

ApontaUA — Collaborative Academic Document Management System

M2 — Lifecycle Architecture | Prepared from the MS1 review and additional source research

1. Related Work

Digital academic materials are distributed through learning management systems, student communities, shared drives and dedicated educational repositories. These approaches make resources available in different ways, but they do not all address the same problem: some focus on teacher-managed course delivery, others on community sharing, and others on professionally authored, reviewed learning resources.

Research on Learning Object Repositories (LORs) identifies quality assurance as an important part of repository design. Clements, Pawlowski and Manouselis reviewed quality approaches used in educational repositories and highlighted instruments such as peer review, user ratings and recommendation systems. Their work argues for treating quality as a process that involves the repository, its contributors and the way resources are evaluated, rather than simply storing files in one place [1]. Earlier work by Clements and Pawlowski also found that trust in resources, organizations and technologies influences how teachers assess and reuse Open Educational Resources (OER) [2].

Open Educational Resources provide a related perspective. UNESCO defines OER as teaching, learning or research materials in the public domain or released under an open licence that permits no-cost access, reuse, adaptation and redistribution [3]. This definition is relevant to platforms that share academic documents, but it is important not to treat every uploaded study document as an OER automatically: the contributor must have the right to share it, and the applicable licence or terms must be clear.

Across these works, several design concerns are relevant to ApontaUA: resources need useful descriptions and metadata; users need effective ways to discover material; content freshness and trust should be visible; and review mechanisms can help users assess quality. These concerns inform the comparison below.

2. Existing Solutions

2.1 Resumos de Informática

During the MS1 analysis, the team reviewed Resumos de Informática, a repository focused on Informatics/Computer Engineering study materials. The presentation identified organization by academic year and subject, and availability of summaries, exercises, slides and other study material as strengths. The team also observed limitations: materials could be outdated, some units no longer belonged to the current curriculum, duplicate materials existed, and the scope was limited to Informatics [9].

This example demonstrates the value of a subject-focused repository, while also highlighting the importance of maintenance, curriculum alignment and content lifecycle management. These findings describe the team's MS1 inspection and should not be interpreted as a new audit of every resource currently available on the website.

2.2 NEI — Apontamentos

The Núcleo de Estudantes de Informática (NEI) at the University of Aveiro provides an online page for student notes. Its public interface offers filters for academic year, curricular year, subject, author and professor, with categories such as bibliography, lecture slides, projects, practical guides and exercises, tests and exams, notebooks and summaries [4]. The page also invites students to contribute materials, currently directing contributions through email [4].

The repository is closely aligned with the local Informatics student community and has a useful academic classification. However, the visible academic-year options on the page reviewed ranged from 2013/14 to 2022/23. That observation does not prove that every resource is outdated, but it shows why publication dates, currency

indicators and active maintenance matter. The public interface does not make a structured, user-facing version history or an explicit validation status for each resource evident.

2.3 Studocu

Studocu is a broad student-contributed platform for study documents. Its help centre describes search filters for institution, course, category, page count, academic year and language [5]. Studocu also describes a document screening process that checks uniqueness and platform criteria; users can request manual review when a document is rejected [6].

This approach is relevant to ApontaUA because it shows how upload workflows can include duplicate screening rather than treating every submission as independent. However, a uniqueness check is not the same as academic validation: it does not by itself establish that a document is correct, current or aligned with a specific course's teaching. ApontaUA therefore intends to distinguish processing/moderation checks from peer or teacher review.

2.4 Moodle

Moodle is primarily a Learning Management System (LMS), where teachers can add resources to their courses. Its documentation lists resource types such as files, folders, books, pages, media areas and URLs [7]. This structure is useful for delivering course material in the context of a specific course or class.

Moodle complements rather than directly duplicates the ApontaUA concept. Its course-based model supports teaching and course administration; ApontaUA is intended to provide a shared collection of materials contributed to and reused across cohorts, with explicit document history, duplicate warnings and a distinct review/validation workflow. Moodle can be configured in many ways, so this is a comparison of its core resource model, not a claim that Moodle cannot support additional features through configuration or plugins.

2.5 OpenStax

OpenStax provides free, openly licensed textbooks written and reviewed by educators. Its higher-education site highlights expert authorship, peer review for accuracy, regular updates and supplementary resources for instructors [8]. These practices illustrate the value of structured editorial review and of maintaining learning content over time.

OpenStax is not a direct substitute for a community repository of local lecture notes: it focuses on curated textbooks and associated teaching resources across subject areas. Nevertheless, its review and update model is a useful reference for ApontaUA's goal of making the status and reliability of materials clearer.

3. Core Features in Existing Approaches

Academic classification: organizing material by course, subject, academic year, category or course space.

Community contribution: allowing students or educators to share documents, although submission workflows differ between platforms.

Search and filtering: helping users locate resources by institution, subject, category and other metadata.

Quality and trust signals: expert review, peer review, ratings, authorship, dates and other indicators that help users assess whether a resource is suitable.

Content maintenance: communicating whether a resource is current and preserving meaningful information when content changes.

Duplicate control: screening similar submissions to reduce repeated content.

Reuse and licensing: clarifying whether and how a resource may be redistributed, adapted or reused.

Research into educational repositories supports the use of peer review, ratings and recommendation approaches as parts of a broader quality strategy [1]. UNESCO's OER recommendation also emphasizes searchability, reuse, quality assurance and suitable open licensing, while making clear that reuse rights depend on the licence and the applicable rights framework [3]. Not every solution implements all these features, and a feature not evident in public documentation should not automatically be interpreted as absent.

4. Comparison of Existing Solutions

The table summarizes features that are publicly documented or that the team observed during MS1. “Not evident” means that the reviewed source did not establish the feature; it does not prove that the feature is unavailable.

Capability	Resumos de Informática	NEI Apontamentos	Studocu	Moodle	OpenStax	ApontaUA (planned)
Main focus	Informatics study materials	UA Informatics community	Multi-institution student documents	Teaching and course delivery	Curated educational textbooks	Shared academic materials at UA
Academic organization	Academic year and subject observed in MS1	Year, subject, author, professor and category filters	Institution, course, category, year and language filters	Resources organized within course spaces	Subject and textbook catalogue	Course → curricular unit → topic → material
Student/community contribution	Materials available; workflow not assessed in MS1	Contributions invited by email	Direct document uploads	Mainly teacher/course-managed resources	Expert-authored and reviewed content	Student and teacher contributions
Quality assurance	Explicit per-document validation not identified in MS1	Explicit validation status not evident on public page	Uniqueness screening and possible manual review	Teacher-managed course context	Expert peer review	Separate moderation and academic review
Duplicate handling	Duplicates observed in MS1	Not evident in reviewed page	Uniqueness screening documented	Not a core resource feature established by reviewed page	Not a primary repository function	Exact-file checks and similar-content warnings planned
Visible version history	Not evident in MS1 review	Not evident in reviewed page	Not established by reviewed sources	Not central to documented resource model	Regular textbook updates; not equivalent to per-upload versioning	Version history planned
Local UA scope	Limited to Informatics	Informatics student community	Broad, with institution/course filters	Depends on course and institution setup	International subject coverage	Designed for University of Aveiro
Reviewed vs unreviewed notes	Not identified in MS1 review	Not evident on public page	Upload screening is not the same as course-level academic validation	Course context provides teacher management, but no shared note lifecycle established here	Textbooks are peer-reviewed	Explicit review/validation status planned

The comparison is based on the MS1 presentation and public documentation listed in References. It is a scoped review, not an exhaustive feature audit or a claim that the proposed combination is globally unique.

5. Identified Gap and ApontaUA's Position

The reviewed solutions confirm that students and teachers already rely on repositories, LMS course spaces and broad study-sharing platforms. The gap identified by the ApontaUA team is not simply the absence of websites containing notes. It concerns the management of these materials as a shared, evolving collection for the University of Aveiro.

A consistent academic hierarchy: course, curricular unit, topic and material, helping users find resources in a predictable location.

A document lifecycle: publication, editing, version history and clear indication of a material's review or validation state.

Collaboration and reuse: allowing students and teachers to contribute to and improve material rather than repeatedly recreating similar documents.

Duplicate awareness: warning contributors about exact or potentially similar materials before publication, while leaving the decision to the user or review workflow.

Trust and discovery: using metadata, dates, search, related materials, ratings and teacher/peer validation to help users assess relevance and reliability.

A University of Aveiro context: focusing on the local academic structure rather than mixing study resources with unrelated services.

These are intended differentiators for the project, not claims that every individual capability is new. Student upload and uniqueness screening already exist in Studocu, while peer review is established in curated educational repositories such as OpenStax. ApontaUA's contribution is the planned integration of these mechanisms with a UA-specific academic structure and a visible lifecycle for community-contributed study materials.

6. Technology Direction

The proposed technology stack is based on the project architecture decisions and is included here to connect the State of the Art to the intended implementation. It is not itself a claim of research novelty.

Area	Proposed direction	Reason
Frontend	React + TypeScript, TanStack Router and TanStack Query	Component-based interface, typed routes and structured API data handling
Backend	Python + FastAPI, modular monolith and REST/OpenAPI	One backend deployment with separated functional modules and documented endpoints
Authentication	Keycloak for MVP; UA institutional login planned for second semester	Enables initial authentication while keeping institutional integration as a later step
Structured data	PostgreSQL	Stores academic hierarchy, material metadata, versions, permissions and review states
Search	PostgreSQL full-text search initially	Avoids a separate search service until a demonstrated need exists
File storage	Persistent storage for original uploads	Preserves original documents independently from metadata and extracted text
Document processing	Python worker for OCR/text extraction and duplicate-candidate analysis, if required for MVP	Keeps long-running work separate from API request handling
Task processing	Celery + Redis, conditional on asynchronous processing being included	Coordinates long-running jobs and their status
Extended search	Elasticsearch only if PostgreSQL search proves insufficient	Defers additional deployment and index-synchronization complexity

The exact file formats, OCR scope and adoption of Celery/Redis still require confirmation against the MVP requirements. These are implementation decisions rather than established findings from the literature review.

7. Conclusion

The analysis shows that academic content sharing is already supported by student repositories, general platforms, LMSs and curated open-textbook collections. Existing approaches offer useful elements such as academic filters, contribution workflows, duplicate screening or expert review. However, they serve different audiences and content lifecycles.

Based on the team's MS1 review and the sources consulted, ApontaUA is positioned as a University of Aveiro-focused collaborative document management system. Its main objective is to combine structured academic organization with contribution, version history, duplicate awareness, search and explicit peer/teacher validation. The value of the project will depend not only on storing documents, but also on making their context, freshness and review status visible and supporting their reuse across cohorts.

References

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[3] UNESCO, “Recommendation on Open Educational Resources (OER),” adopted 25 November 2019. https://www.unesco.org/en/legal-affairs/recommendation-open-educational-resources-oer

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[8] OpenStax, “Higher Education — Free college textbooks, peer-reviewed.” https://openstax.org/higher-education

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